

Kauda Baji: Proposed Rules

For review — July 2026

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Why this document

I'm currently working on a hobby project to build an AI agent that plays Kauda Baji. The idea is to build a complete model of the game in code, then design AI agents that can play it. In the future, time permitting, there may even be a robot that plays the game with you in real life.

I've progressed though making models for the board, player, pieces, and moves. The next step is defining the rules. This must be completed before moving to designing the AI agent.

And this is where I need your help!

The ask is for you to please read this document and provide feedback if you agree or disagree with each rule. Because a computer will be playing using these rules, they must be fully unambiguous.

I would be grateful for your time and feedback.

If you'd like to follow my progress, the code lives at <https://github.com/MGNaik/KaudiPandit>.

Terms used

Our game is played in Gujarati, but code is typically written in English so I've had to come up with some terms for common concepts. Here are the terms I will use in this document, and their meanings.

Piece

One of 6 movable objects belonging to one player.

Tuple

A player's own pieces stacked on the same square and moving together as a unit.

Tuple Size

The number of pieces in a tuple. A tuple of size n is called an n -tuple — a 1-tuple is a single piece on its own, a 2-tuple has size 2, a 4-tuple has size 4, etc.

Kaudi

One of six cowrie shells thrown together to generate a move. Plural: *kaudis*.

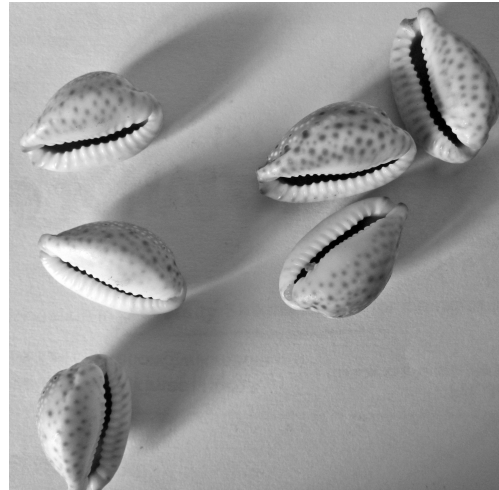
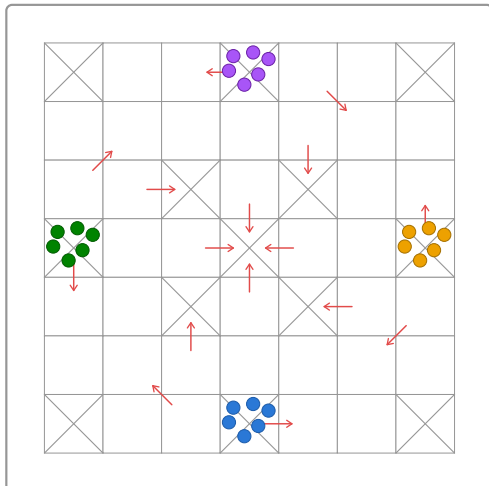
Safe square

A square marked with a corner-to-corner cross. The starting squares and the centre square are safe.

Kill

Removing an opponent's tuple from the board. Killed pieces go back to their owner's starting square and begin their journey again.

The board and pieces



The board is a 7×7 grid of squares. Four players sit at the four edges; each player owns the edge nearest them.

Moves are generated by throwing a set of six *kaudis* (cowrie shells), and the outcome of the throw is used to move the players' pieces around the board.

Each player has six pieces of their own colour. Every player's pieces start on the same square: the crossed square in the middle of that player's own edge. This is the player's **starting square**. From there, pieces travel around and inward, and all four players finish on the single square at the very centre of the board.

The objective of the game is to get all six pieces from the starting square to the centre square. The first player to do so wins.

The rules

Rule 1 — Determining the first player

Before play begins, every player throws their kaudis once. Whoever rolls the highest value goes first. Play then proceeds **anticlockwise** around the table from there.

Example: Red throws 4, Blue throws 6, Green throws 2, Yellow throws 12. Yellow goes first (12 beats everything), then play continues anticlockwise from Yellow's seat.

Example: Red throws 5, Blue throws 5, Green throws 6, Yellow throws 3. Green goes first with the single highest value, 6.

If two or more players tie for the highest value, only those tied players throw again to break the tie; whoever wins that re-throw goes first. Players who weren't tied are not involved in the re-throw.

Example: Red throws 6, Blue throws 6, Green throws 4, Yellow throws 2. Red and Blue are tied for highest and throw again, just the two of them, until one beats the other.

Rule 2 — Throwing the kaudis

A turn begins with a throw of all six kaudis. The value of a throw is the number of kaudis that land mouth-up, except that a throw with **no** kaudis mouth-up is worth **12** rather than 0. So a single throw is worth 1, 2, 3, 4, 5, 6, or 12.

A throw of 6 or 12 earns the player an extra throw, and the value of that extra throw is **banked** alongside the first. If a player throws 6 or 12 three times in a row, they forfeit the rest of their turn — none of the banked values may be used.



3 mouth-up → value 3



6 mouth-up → value 6 (bonus: throw again)



0 mouth-up → value 12 (bonus: throw again)

Example: A player throws 6 (a bonus), throws again and gets 4 — since 4 isn't 6 or 12, the turn stops there with two values banked to spend: 6 and 4. A player who instead throws 6, then 12, then 6 has hit a bonus three times in a row and forfeits the whole turn — none of those three values may be used.

Rule 3 — Using a throw to move

Each throw value produced during a turn moves one piece that many squares. It is the player's decision which pieces they allocate throw values to.

If a turn produces more than one value because it included one or two throws of 6 or 12 (see Rule 2), the total value may be spent on a single piece (added together), or spread across several different pieces, in any combination the player chooses.

However, a single throw value cannot be split between two pieces. A throw of 6 and 3 in the same turn, for example, can move one piece 9 squares in total, or two different pieces 6 and 3 squares each, but the 6 and the 3 can never themselves be divided further.

Throws that can legally be used by any of the player's pieces must be used, there is no option to forfeit or skip other than two special cases in Rule 9 and Rule 16.

Example: A player throws 6 (a bonus, so they throw again), then 12 (another bonus, so they throw once more), then 4 — not a 6 or 12, so the turn stops there with three values banked: 6, 12, and 4. Legal ways to spend them include: one piece moves 22 (6+12+4); one piece moves 18 (6+12) and a second piece moves 4; or three different pieces move 6, 12, and 4 respectively. Not legal: splitting the 6 into a 4-and-2 to feed two pieces.

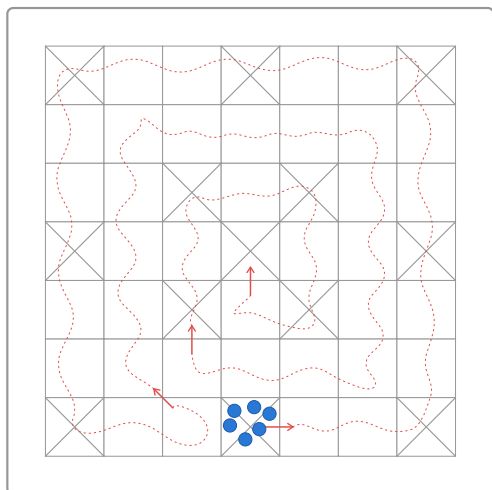
Example: A player throws 5 on their first throw. Since 5 is neither 6 nor 12, no bonus throw follows — the turn produces just that one value. The whole 5 must go to a single piece; there's nothing to spread, since there's only the one value.

Rule 4 — The movement path: anticlockwise out, clockwise in, three arrows to the centre

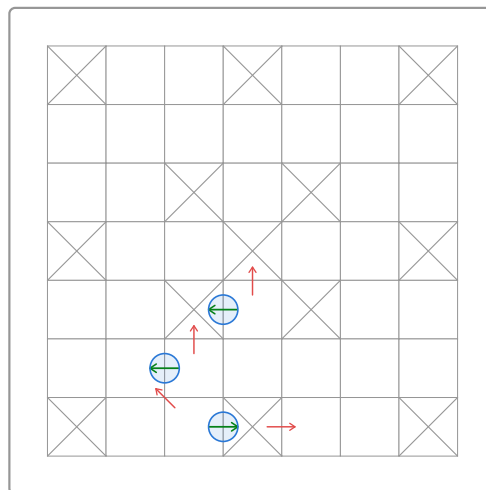
Every piece follows the same route from its starting square to the centre. The route is identical for all four players, just rotated to begin at each player's own edge.

A piece begins on its **starting square** — the crossed square in the middle of the player's edge — and travels anticlockwise around the outer ring of the board. At the **first arrow**, it turns inward onto the first inner band and continues **clockwise** around that band. At the **second arrow**, it turns inward again onto the second inner band, still travelling clockwise. The **third and final arrow** then carries the piece onto the centre square, completing its journey.

At each of the three arrows, a piece is not forced inward: its owner may choose to follow the arrow onto the next band, or keep the piece moving straight ahead on its current band instead, for strategic reasons. (The first arrow carries one further condition — see Rule 5.)



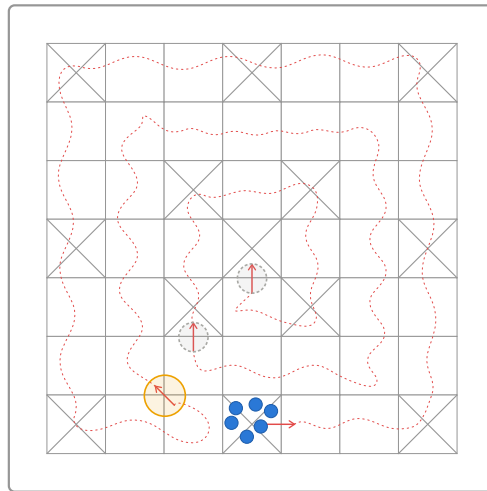
The full route: anticlockwise on the outer ring, clockwise on each inner band, three arrows carrying the piece to the centre.



Turning at each arrow is optional — shown on the bottom player's path; the same three points apply, rotated, for every player.

Rule 5 — The Markut rule: a kill is required to enter the first inner band

A player may not turn any piece inward at the **first** arrow until that player has killed at least one opposing piece, at any point earlier in the game. This is a one-off unlock, not a per-piece cost: once a player has made their first kill, *all* of that player's pieces may enter the first inner band freely from then on. No kill is required to enter the second inner band or the centre — the requirement applies only to the first arrow.



Arrow 1 is gated by this rule (highlighted yellow); arrows 2 and 3 (dashed grey rings) are not.

Rule 6 — A kill earns a bonus throw

Making a kill — landing on and killing an opponent (Rule 11 or 13), or killing on exit (Rule 12) — earns the killer an immediate extra throw. Besides this and throwing 6 or 12 (Rule 2), nothing else grants a bonus throw.

A bonus throw earned from a kill does **not** count towards the limit in Rule 2 of forfeiting after rolling 6 or 12 three times in a row — that limit is only about consecutive 6s and 12s *from the kaudis themselves*. A kill's bonus throw doesn't reset or extend that count either way.

Rule 7 — Tuples form only when a player chooses to move pieces off together

When one of a player's pieces or tuples lands on a square already holding one or more of that **same** player's own pieces or tuples, nothing merges automatically — they simply share the square as separate pieces, exactly as before. A tuple only forms at the moment its owner chooses to move two or more of them off that square **together**, as a single unit; forming and moving happen in the same action, and that move must obey the divide rule (Rule 9) for the combined size. If the pieces are instead moved off individually, or not moved together, no tuple forms.

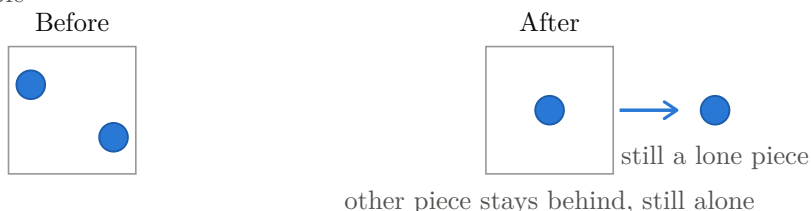
Once formed, a tuple moves, and is killed, together as one unit (see *Tuple* in Terms used). Tuple size is a plain sum of the pieces merged into it, with no cap other than a player's own six pieces — two dugalu (2-tuples) can combine into a chogalu (4-tuple), and so on up to a single 6-tuple holding all of a player's pieces at once.

A tuple can form on any square other than a player's own **starting square** — ordinary squares and safe squares alike. The starting square is the one exception: pieces waiting there remain single pieces no matter how many share it, and can never be moved off together as a tuple from there. A tuple can only begin forming once a piece has left the starting square.

Moved together: tuple forms



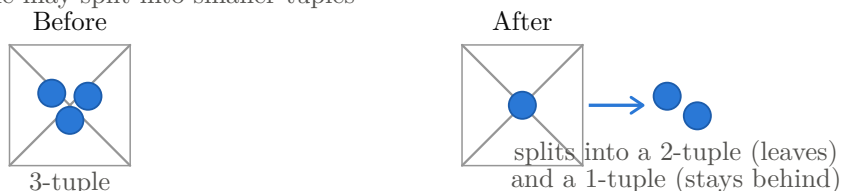
Moved separately: no tuple



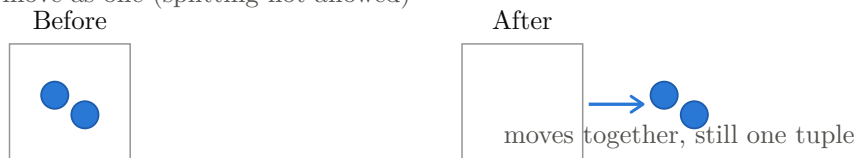
Rule 8 — Tuples may only dissolve on a safe square

A tuple may split, but only while it occupies a safe square. The split need not go all the way down to single pieces — a tuple may divide into any combination of smaller tuples and pieces, with some parts leaving the square and others staying behind. Everywhere else on the board, a tuple must move and be killed as a single unit until it next reaches a safe square.

On a safe square, a tuple may split into smaller tuples



Off a safe square: must move as one (splitting not allowed)



Rule 9 — Moving a tuple: the divide rule

A tuple of size n can only be moved using a throw value that n divides without remainder. A 2-tuple needs a throw of 2, 4, 6, or 12; a 4-tuple needs a throw of 4 or 12; a lone piece (a 1-tuple) can move on any throw, since 1 divides everything. A throw value that doesn't divide evenly by a tuple's size simply cannot be used to move that tuple.

When a tuple *does* move, it doesn't advance by the full throw value — it advances by the throw value **divided** by its size. A 2-tuple moved with a throw of 6 advances 3 squares ($6 \div 2$), not 6; a 4-tuple moved with a throw of 12 advances 3 squares ($12 \div 4$). The bigger the tuple, the less distance the same throw actually covers.

If the player has no usable throw values under this rule, they forfeit the throw value and their turn is over.

Tuple size	Usable throw values	Squares moved
1-tuple (lone piece)	1, 2, 3, 4, 5, 6, 12 — any throw	same as the throw
2-tuple	2, 4, 6, 12	1, 2, 3, 6
3-tuple	3, 6, 12	1, 2, 4
4-tuple	4, 12	1, 3
5-tuple	5 — only an exact throw of 5 works	1
6-tuple	6, 12	1, 2

Rule 10 — Safe squares: no kills, ever

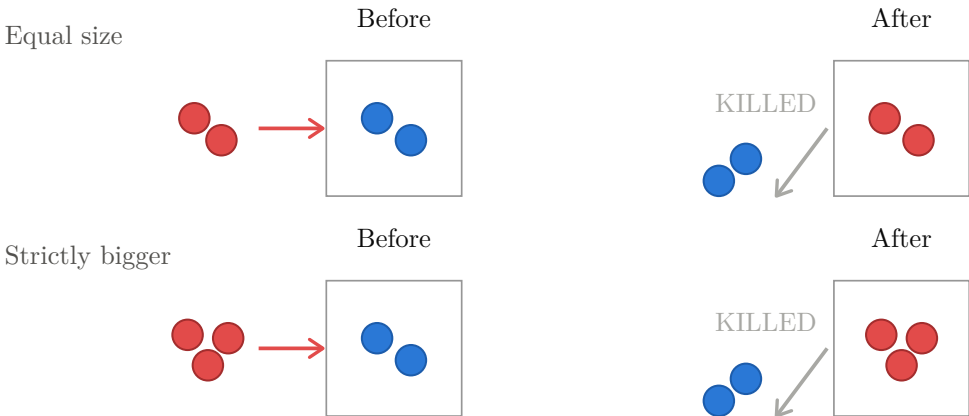
Twelve squares carry a corner-to-corner cross, and one more sits at the centre — these are the **safe squares**, marked on the board. Every other square is ordinary.

No kill of any kind can happen on a safe square. Any number of tuples, of any sizes, from any players, can share a safe square. Landing there kills nothing, and leaving a shared safe square kills nothing.



Rule 11 — Landing on an opponent: equal-or-bigger kills, smaller coexists

If your tuple lands on a square holding an opponent's tuple, and your tuple is the **same size or bigger**, the opponent's tuple is killed. Your tuple takes the square; their pieces go back to their start.



The same happens if the landing tuple is strictly bigger — a Red 3 landing on a Blue 2 kills it just the same. If instead your tuple lands on a square holding an opponent's tuple and yours is **strictly smaller**, nothing is killed. Both tuples share the square.



Notice what this means: whenever two opponents share a square, the tuple that was there first is always the bigger one.

Rule 12 — Leaving a shared square: the bigger tuple kills on exit

When two opponents share a square and the **strictly bigger** tuple moves away, it kills the smaller tuple as it leaves. (This rule is for exactly two parties sharing a square, each with a single tuple — see Rule 14 if that one opponent has multiple separate tuples there instead, or Rule 15 when a third or later opponent is involved.)



The reverse is safe: if the *smaller* tuple moves away first, no kill happens — it simply escapes.



The smaller tuple sits under threat until either it leaves, or the bigger tuple leaves and kills it on the way out.

Rule 13 — Landing on an opponent with multiple separate tuples: the largest one you can beat dies

An opponent can have more than one of their own tuples sitting on the same square without them being merged (see Rule 7) — for example a 1-tuple and a 3-tuple side by side, never moved together. When your tuple lands on that square, look at all of their tuples that are of **equal or lesser size** than yours — the ones you're big enough to beat — and kill only the **largest** of those. Every other tuple of theirs is safe: any of theirs smaller than the one you killed are spared, and any of theirs bigger than yours were never at risk to begin with.

Lander size 3: kills the 3-tuple (the largest one it can beat)



Lander size 2: kills the 1-tuple instead (3-tuple ineligible)



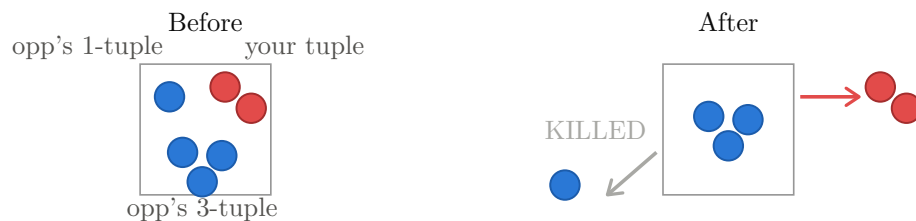
This is still a **two-player** interaction — one opponent owning several tuples, not several different opponents (Rule 15 covers that case). This rule is specifically about **landing** — see Rule 14 for what happens when someone leaves a square like this instead.

Rule 14 — Leaving a square where multiple tuples remain: the same largest-you-can-beat logic

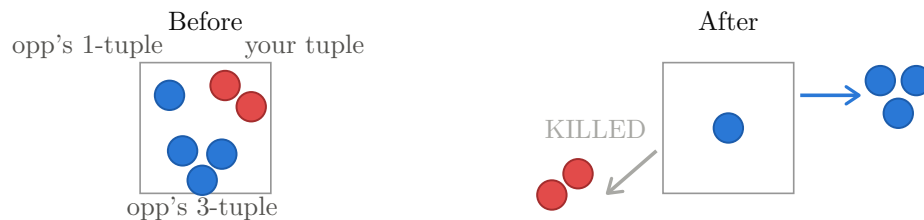
When a tuple leaves a square, it can kill the **largest** of the **opposing** tuples left behind (belonging to a different player) that is **strictly smaller** than the leaving tuple — matching Rule 12, where only a *strictly bigger* tuple kills on exit (a tie never kills). Every opposing tuple that's the same size as the leaver or bigger is safe, and any opposing tuple smaller than the one killed is spared too. This applies no matter **which** tuple leaves — yours, or one of an opponent's own separate tuples — and no matter how many tuples are left behind.

This is a generalisation of Rule 12, not a separate mechanic: with only one tuple left behind, “the largest remaining tuple strictly smaller than the leaver” is just that one tuple if it qualifies, so Rule 12 is the two-tuple special case of this rule.

Your tuple (size 2) leaves: their 3-tuple is safe (bigger), kills their 1-tuple instead



Their 3-tuple leaves instead: kills YOUR tuple (their 1-tuple was never a target)



Rule 15 — Three or more players on one square: no kills at all, landing or leaving

Rules 11–14 are only for a square shared by exactly **one** opponent (whether that opponent has one tuple there or several unmerged ones). Once a square already holds tuples from **two or more different** opponents, everything

changes, for **both landing and leaving**: **no kill can happen there, regardless of size**, in either direction.

Landing: a tuple landing on such a square simply joins the pile, no matter how big or small it is compared to what's already there.

Lander bigger than both existing: still no kill



Lander smaller than both: no kill either



Leaving: if a tuple currently shares a square with **two or more** other opponents' tuples and moves away, it kills nobody on exit either. Exit kills (Rule 12) only ever happen when exactly one opponent's tuple is present.

Rule 16 — Reaching the centre: exact throw required

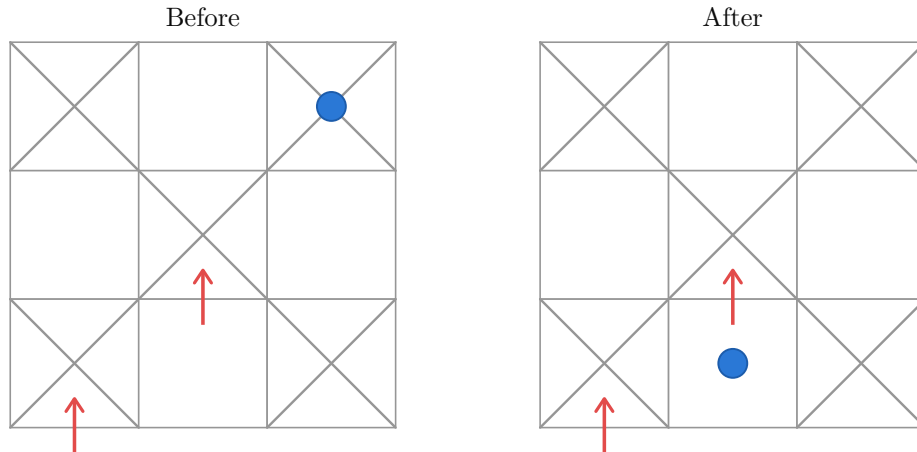
Sending a piece or tuple on to the centre square, and thus off the board, requires an **exact** throw value which lands the piece or tuple onto the centre square.

As an outcome of this rule, a number of notable cases arise:

1. Player throws a value smaller than the required value to land the piece or tuple on the centre square

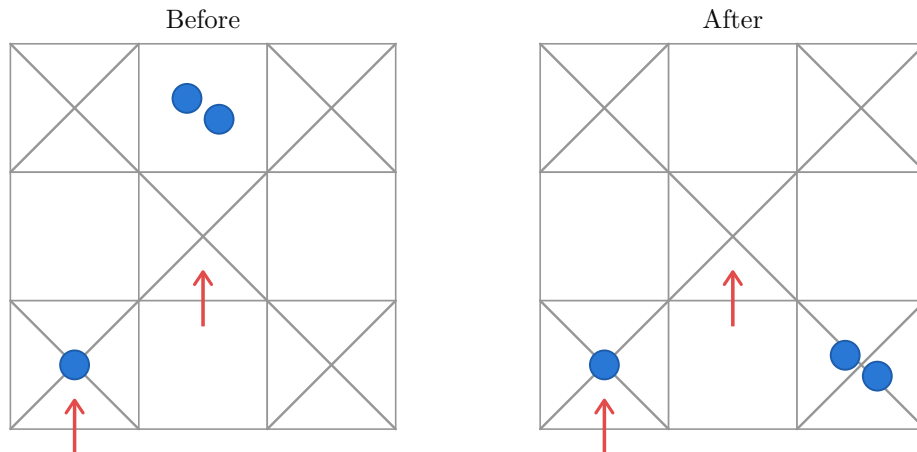
In this case, the player doesn't have the roll value to land the piece/tuple on the centre. However, the lesser move is still an ordinary, legal move, and must be used as per Rule 3.

Example: A single piece is 4 squares away from the centre and the player has no other pieces on the board. A throw of 3 falls short of landing the piece onto the centre square, but the move is still legal. Since no other piece on the board can use that 3, this piece must advance with it.



This turn produced 3. The piece is 4 away, so the 3 does not reach the centre — but the move is still legal, and must be used: the piece advances to 1 away.

Example: A 2-tuple is 5 squares away from the centre and a single piece is 8 squares away. A throw of 6 falls short of the 10 needed to land the 2-tuple or the 8 needed to land the single piece on the centre, but the throw is still usable. The player must use the throw value and decide which of their pieces or tuples to move.



This turn produced 6, which moves this 2-tuple 3 squares ($6 / 2$, per Rule 9) — short of the 5 needed to land. One legal choice: the 2-tuple advances 3 squares, to 2 away.

2. Player throws a value larger than the required value to land the piece or tuple on the centre square

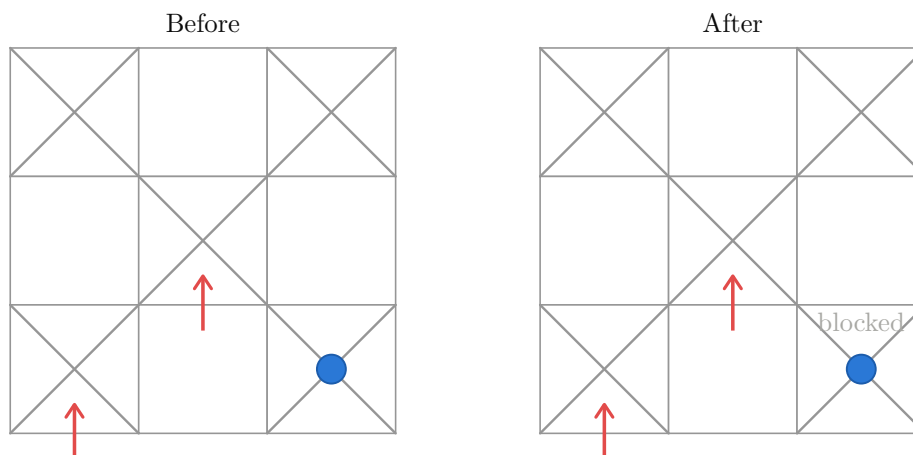
In this case, the player has a roll value that is larger than that required to land the piece or tuple on the center. The larger throw value cannot be used to land the piece or tuple on the centre square.

Instead, the player has these options, and is free to choose between whichever of them are available — neither is forced over the other:

- If a different one of the player's pieces or tuples can legally use the throw value without overshooting, move that piece or tuple with it instead.
- Move the piece or tuple around the inner band again instead of turning in to the centre square, as per Rule 4. This is always a legal, ordinary move, whether or not any other piece can use the value.

If no different piece or tuple can use the value — so option (a) isn't available — the player is not obliged to take option (b) either, and may instead forfeit the throw value and finish their turn.

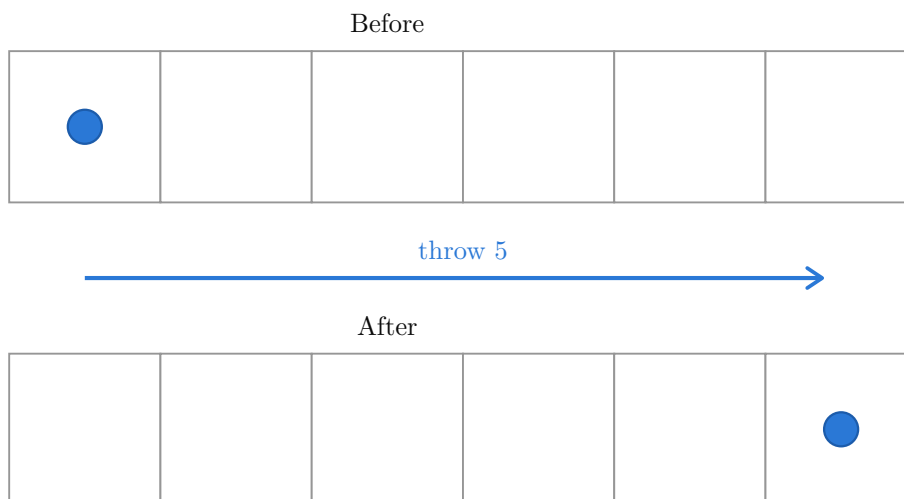
Example: A piece is 2 away. A throw of 5 is too big to land it — it simply cannot be used to move this piece onto the centre.



This turn produced 5. The piece is 2 away, so the 5 overshoots the centre — it simply cannot be used to move this piece there.

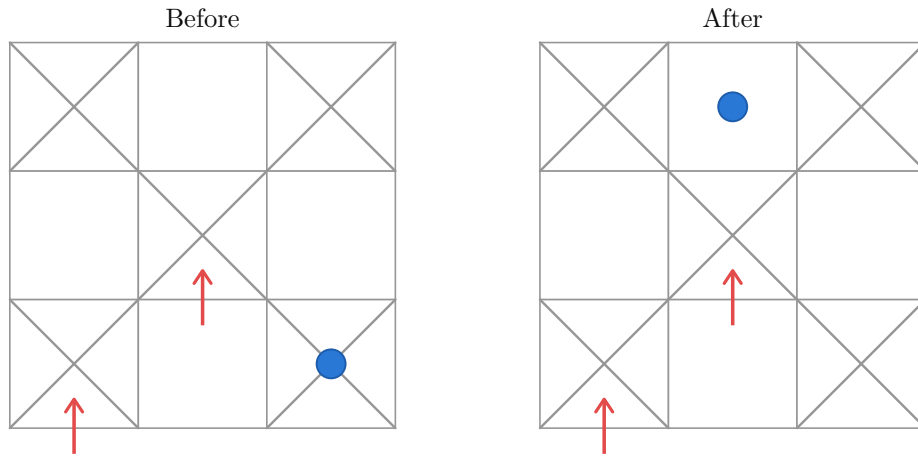
Example — option (a): With that same throw of 5, if a different piece elsewhere on the board can legally move 5 squares, the player may choose to use it there instead.

A different piece, elsewhere on the board



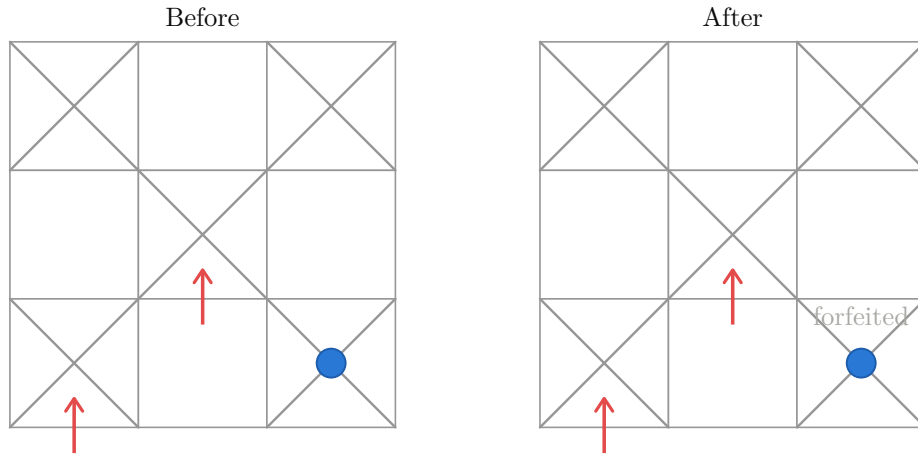
Option (a): the 2-away piece near the centre cannot use the 5 to land, but this piece can move 5 squares along its ordinary path — the player may choose to send the 5 here.

Example — option (b): Alternatively, the player may spend the 5 on the 2-away piece itself: declining the arrow (Rule 4), it loops around its inner band and ends up 5 away from the centre, waiting for an exact value on a later turn.



Option (b): the 2-away piece spends the 5 on itself — declining the arrow (Rule 4), it loops around its inner band and ends up 5 away, waiting for an exact value later.

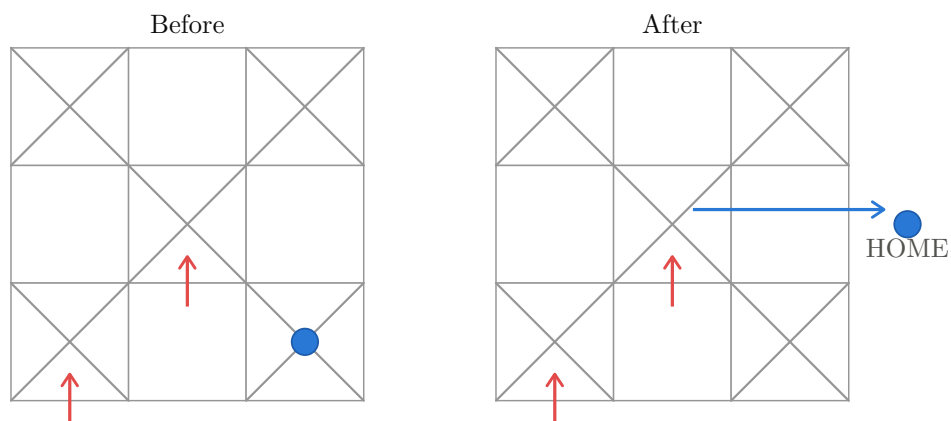
Example — forfeit: If instead no different piece anywhere can use the 5 (option (a) unavailable), the player isn't obliged to take option (b) either — they may simply forfeit the value and end their turn, the same way values are lost when a triple bonus forfeits a turn (Rule 2).



No different piece can use the 5, so option (a) is unavailable. The player need not take option (b) either — they may forfeit the 5 and end their turn.

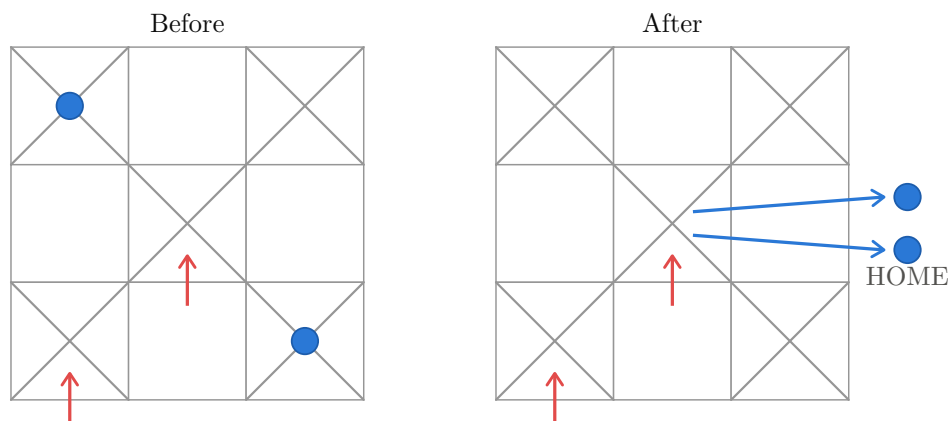
A throw of 6 or 12 always earns an immediate extra throw (Rule 2), even when that 6 or 12 itself overshoots and can't bring a piece home. A later throw in that same chain — alone or summed with earlier ones — can still complete the piece's journey if it lands exactly; if instead it falls short, it is not forced onto this piece and follows the same cases above.

An exact throw lands the piece on the centre — and off the board



This turn produced 2 (not a 6 or 12, so no bonus throw follows). The piece is exactly 2 away, so the 2 lands it precisely on the centre, taking it off the board for good.

Two pieces close enough: the player chooses which value goes where



A throw of 6 earns a bonus throw of 2, so this turn produced 6 and 2 — and both take a piece off the board via the centre. The player is free to assign either value to either piece.

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